

Diagnose Before You Inspect: Failure Attribution Prescribes a Deterministic Symbolic–Neural Critic for Slides

Anonymous submission

Abstract

Vision–language models (VLMs) routinely pass slides with visible defects, but a failed inspection does not identify why the model failed. We test whether separating failure mechanisms can determine the appropriate inspection method. Using a structured-geometry oracle that is lossless with respect to the declared IR, we distinguish defects that fall below the models’ effective perceptual threshold, defects that are visible but suppressed by a whole-rubric prompt, and *reference-assisted* defects that become resolvable only when a clean twin is available. The resulting critic assigns declared geometry and terminology to symbolic linters, format-suppressed defects to atomic VLM queries with required evidence, reference-assisted defects to paired comparison, and page semantics to a finetuned examiner. On three vendors, atomic elicitation outperforms a compute-matched self-consistency baseline with a single call. The deterministic composition reaches 0.957 macro balanced accuracy on held-out validation. A later frozen check on a disjoint set scores lower when unresolved reference requests count as misses. Clean controls for the reference-assisted class, pixel-recovered structure, and real SLIDEAUDIT layouts identify the conditions under which this assignment transfers.

1 Introduction

Slide- and document-generation agents need critics that can locate layout and content defects before an output is returned or revised. A common choice is a general-purpose vision–language model (VLM) prompted to grade the rendered slide against a rubric. Yet the resulting judgment does not reveal why a defect was missed. For example, a strong VLM may answer “no defect” when a title visibly spills out of its box. The same answer is consistent with two different failures: the model may not resolve the overflow, or the prompt may fail to elicit a signal the model can use. These explanations imply different remedies, ranging from better visual input to a different query or a symbolic tool.

This distinction becomes more important when defects vary in scale and in the evidence required to identify them. Some defects are *sub-perceptual* under the tested protocol, such as a three-pixel alignment offset or a one-point font mismatch below the models’ effective acuity. Changing the prompt, scale, or resolution does not recover them. Other defects are *format-suppressed*: the model detects them

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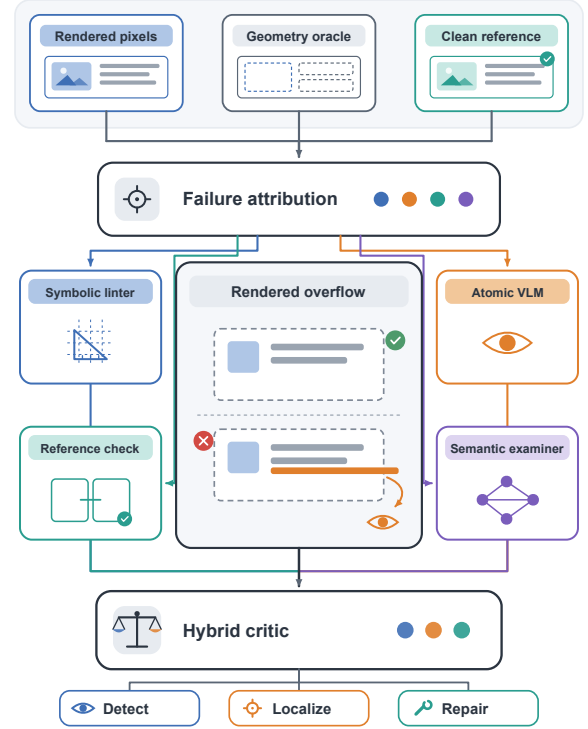


Figure 1: **Diagnose before inspection.** A geometry oracle attributes each failure, then routes it to symbolic linting, atomic VLM elicitation, reference comparison, or semantic examination for deterministic fusion by the hybrid critic.

when asked one targeted question with required evidence, but misses them when one pointwise call must score the full rubric. A third set is *reference-assisted*. These defects become resolvable when the model can compare the slide with a clean twin, but not from the single slide alone. Treating all three outcomes as generic failures of “visual understanding” obscures which intervention is appropriate.

We measure this distinction with an attribution protocol based on a structured-geometry oracle that is lossless with respect to the declared IR. Each slide has an intermediate representation (IR) containing element bounding boxes, text, and style. We present matched defective and clean slides as

an image, the structured oracle, a VLM-generated caption, or the image together with the oracle. We cross these inputs with detection, localization, and repair tasks. Unlike a model-generated caption, the oracle exposes the declared structure without first passing it through another visual model. Success with the oracle after failure on the image therefore operationally identifies a perception bottleneck at the tested interface. Failure with both inputs indicates that changing the neural input alone does not solve the class.

The attribution results determine a fixed division of labor (Fig. 1). Symbolic linters inspect declared geometry and terminology. A language model handles text and structural semantics. VLMs inspect perceivable defects under the elicitation format supported by the diagnosis, using either an atomic query or a clean reference. We also construct G7, a render-containment overflow whose declared box is legal although its rendered content crosses the boundary. This class tests the part of the design that a structure-only linter cannot observe. The complete symbolic–neural critic assumes an IR-owning generation agent. When only third-party pixels are available, the linter lacks its required input and the available critic reduces to the re-elicited VLM (Sec. 7).

The paper makes four connected contributions. First, the structured-oracle protocol separates sub-perceptual, format-suppressed, and reference-assisted failures, and we repeat the attribution on real CC-licensed layouts (Secs. 4 and 7). Second, defect-specific elicitation recovers signals suppressed by a whole-rubric prompt, whereas G1 text overflow and S6 image/text contradiction still require a clean reference (Sec. 5). Third, a compute-matched control shows that this recovery is not explained by additional test-time compute: for G7, one atomic call exceeds ten-draw self-consistency by 0.20–0.31 across three vendors. Finally, the resulting deterministic composition, which uses no learned assignment model, reaches 0.957 macro balanced accuracy on held-out validation. A later frozen image-arm check reaches 0.826 on a disjoint set when unresolved reference requests count as misses. Clean-control errors, failed structure recovery, and transfer to real layouts define the scope of these results.

2 Related work

The VLM perception bottleneck. Prior work separates several sources of visual failure. Chart understanding can be limited either by the vision encoder or by extraction from its representation (Liu et al. 2025b). Direct readouts likewise show that visual information available in an encoder may not be used effectively by the full VLM (Fu et al. 2025). MMVP constructs image pairs that expose systematic shortcomings associated with CLIP-based visual representations (Tong et al. 2024). We study quality inspection of generated artifacts, where declared geometry provides a structured observation of the visual content and the attribution result can be used to choose the critic’s inspection method.

Perception/reasoning attribution. Concurrent work uses natural-language descriptions to separate perception from reasoning in abstract visual-reasoning benchmarks (Wang et al. 2025). Our setting instead provides a structured repre-

sensation of boxes, text, and style, together with exact labels from controlled injection. The representation avoids using a caption as the oracle and lets the attribution select a downstream inspection method. The LED benchmark also injects empirically grounded structural errors, but it evaluates errors in document-layout-analysis predictions and studies limitations of overlap-based metrics (Heo et al. 2025). We inject errors into rendered slide artifacts and use the resulting diagnosis to compose a critic.

Relative vs. absolute judgement. VLM judges can preserve rankings while producing uncertain absolute scores (Kumar et al. 2026). Pairwise assessment has also improved zero-shot NLG evaluation for some LLMs, although it exhibits positional bias (Liusie, Manakul, and Gales 2024). Protocol choice can introduce other biases as well: pairwise judgments may be more sensitive than absolute scores to distractor features (Tripathi et al. 2025). Our experiments use relative and atomic queries as diagnostic interventions rather than assuming that either is generally superior. The useful result is their class boundary: they recover G1 text overflow and S6 image/text contradiction, but not fine G3 alignment or S3 terminology drift.

Slide generation, evaluation, and reward. Recent slide systems combine generation with evaluation or revision. AutoPresent studies programmatic slide generation and iterative self-refinement (Ge et al. 2025). AeSlides optimizes layout with verifiable aesthetic rewards (Pan et al. 2026), while EvoPresent uses an aesthetic model for scoring, adjustment, and comparative feedback (Liu et al. 2025a). VLM-SLIDEVAL tests element extraction, perturbation sensitivity, and narrative ordering, and argues for calibrated critics in iterative pipelines (Kang, Bao, and Goswami 2025). SLIDEAUDIT contributes an expert-derived defect taxonomy and annotated slides for evaluating design critique (Zhang et al. 2025). We use its taxonomy and real slides for transfer evaluation, then audit five published scorers and one closed VLM judge on the constructed render class (Sec. 5.3).

3 Problem setup and defect taxonomy

Task and representation. The critic inspects a single rendered slide, optionally with its declared structure, and must *flag and localize* defects of a fixed set of types; a detection counts only when it fires on a defective slide and stays silent on its matched clean twin. Every slide is an intermediate representation (IR) of typed elements with bounding boxes, text, and style; we inject defects programmatically into the IR and render, yielding exact labels and matched (defective, clean) pairs. The taxonomy (Table 1) has a *geometry* group G1–G6, a *semantic* group S1–S6, and one constructed render class G7; it is aligned to the expert-derived SLIDEAUDIT taxonomy (Zhang et al. 2025) via a bidirectional map (crosswalk in the supplement), and results are reported in its canonical classes. We add G7 to test the linter’s blind spot: the declared box is legal (inside the safe margin, no overlap) but the rendered content overflows its container, so the defect exists *only* in pixels. On real agent-generated decks G7 is common, affecting most slides of a Zenodo10K slice and

Table 1: Defect taxonomy. The geometry group is detected symbolically; the semantic group is the examiner’s domain; G7 is constructed to be linter-blind. “Channel” is the engine the diagnosis assigns (Sec. 4–5).

Group	Class	Description	Channel
Geometry	G1 text overflow	text exceeds its box	VLM+linter
	G2 element overlap	boxes intersect	linter
	G3 alignment offset	small mis-alignment	linter
	G4 font-size inconsistency	off-grid font size	linter
	G5 brand-color violation	off-palette color (CIELAB color difference ΔE)	linter
	G6 margin violation	element past safe margin	linter
Semantic	S1 title/body mismatch	body contradicts title	VLM
	S2 narrative-order break	deck sections out of order	LLM
	S3 terminology inconsistency	term drift across deck	term linter
	S4 density violation	slide over/under-packed	LLM
	S5 missing logic section	dropped argument step	LLM
	S6 image/text contradiction	figure contradicts caption	VLM
Render	G7 containment overflow	legal box, rendered overflow	VLM (atomic)

over half of an AutoPresent slice.¹

Attribution modalities and metric. To attribute a failure, we define four input conditions. Condition A contains only the rendered image. Condition B contains the structured oracle, which preserves element boxes, text, and style from the source IR without loss. Condition B’ contains a VLM-written caption and serves as a lossy control. Condition C combines the image and oracle. Under each condition, we ask the model to detect, localize, and repair the defect. The attribution reads off the A-vs-B contrast: if the model fails on the image alone (A) but succeeds given the oracle (B), we call the class *perception-bottlenecked* operationally: the task becomes solvable when the same information is made explicit in structure that is lossless with respect to the declared IR. This is a diagnosis of which engine can solve the class, not a claim about the model’s internal representations. If it fails under both, the class is unsolved even at the structured interface, and we assign it to a symbolic method.² We report *balanced accuracy on the paired clean controls*: a detection counts only if it fires on the defective slide and stays silent on its matched clean twin, so a model cannot score by flagging everything.

Two linters. A geometry linter detects G1–G6 from coordinates and rules (brand color via CIELAB ΔE_{2000} , not RGB distance), at zero false positives on clean slides. A terminology linter handles S3 from an occurrence table plus near-duplicate clustering, image-free. Both are the “detectors of record” prescribed by the diagnosis.

4 Diagnosis: two kinds of blindness

Pointwise image-only inspection misses fine geometry. Under the conventional setup, one slide receives one rubric score with no reference slide for comparison. We call this *pointwise image-only* inspection. In this setting, G2–G6 sit

¹93-deck Zenodo10K slice: 81/93 decks affected (857/3167 legal boxes); 14-deck AutoPresent slice: 8/14 decks (20/50 legal boxes).

²The oracle’s ground-truth labels are stripped from the model’s view by a regression-tested de-leaking pass, so B tests perception of structure, not label leakage. B’ is a control, not a second oracle: a caption re-injects the perceptual error it is meant to factor out, so B’s advantage is by construction.

at chance for 4B and 8B models, and a 30B breaks only G1 overflow. The floor remains under the tested changes: swapping the vision encoder across *five* families (SigLIP2, an LLM-based encoder, InternViT, a native-resolution NaViT, MoonViT) leaves every model at the random line (differing only in abstain-vs-over-flag bias), and raising resolution 1536→2048 changes nothing. The lever is elicitation and magnitude, not encoder or pixel budget, so the floor is a property of the *pointwise* protocol, not a hard capability limit. For declared geometry, the symbolic linter reads coordinates directly and reaches balanced accuracy 0.75–1.0 with zero false positives where the IR is available.

Some apparent perceptual failures depend on elicitation.

The same setting hides a second, opposite phenomenon. For G1 overflow and S6 image/text contradiction, pointwise balanced accuracy is 0.50, indistinguishable from the sub-threshold tail, yet a two-alternative forced choice between the defective and clean slide jumps to 1.00: the defect is perceivable, but the single-slide pointwise call does not surface it. We therefore treat G1/S6 as *reference-assisted*: recovery needs the clean twin, not just a renamed single-image question. G3 alignment behaves differently, but only *above a magnitude threshold*: sweeping the injected offset as an internal contrast (one element shifted from its aligned sibling column), the forced choice saturates to 1.00 by 16 px (0.83% of slide width) while the sub-threshold tail (≤ 8 px) stays at chance even side by side. So fine alignment is not a class-level capability floor: above threshold it is format-suppressed and recoverable, with a sub-perceptual residue only in the small-offset tail. The connecting thread is a *magnitude-gated* recoverable-versus-sub-perceptual boundary rather than a clean geometric/semantic split, and S3 terminology (below) remains the one defect no elicitation rescues.

On the rebuilt corpus we also now have a human spot-check on the actual pairs cited here, not just the earlier diagnosis runs. Replacing only the stale legacy G3/G5 samples and keeping the other 55 pairs fixed, the v2 pass verifies every sampled defect as visible and every clean twin as clean (73/73, Wilson lower bound 0.95 for both), with source-structure faithfulness 55/55 on the auditable classes. The delta is exactly where the re-operationalisation says it should be: G3 shifts from 0/7 on the old external-reference sample to 8/8 on the corrected within-list misalignment sample, and G5 from 0/7 to 10/10 on the corrected within-list chromatic clash sample; the non-replaced 55 pairs are unchanged.

Terminology remains unresolved. Not every consistency defect is format-suppressed. S3 terminology inconsistency is *not* rescued by forced choice (0.625, heavy position bias): reading the same term across deck pages and spotting a subtle variant is below threshold even side by side. The structured-oracle channel, which feeds the model the full deck text with both variants present, also fails (0.56): with both variants already explicit in the input, this failure cannot be OCR, and instead reflects cross-page term-matching that does not survive the yes/no framing. The image channel is consistent with fine-grained glyph discrimination and the oracle channel fails even with the text in hand, so neither neural method is

reliable. S3 is therefore assigned to a symbolic terminology linter, which reaches balanced accuracy 1.000 against the VLM’s 0.69, a negative result that changes the prescribed method.

The G6 page-offset boundary depends on magnitude. Re-posing a defect as a within-slide contrast does not rescue every class. It fails outright on G6. We redefine G6 margin as a *page offset*: the whole content block is shifted toward one edge (every element together, so internal alignment is preserved and the defect cannot be misalignment), leaving one margin crowded against (or running off) the edge and the opposite side empty.³ Across all six VLMs and both pointwise framings the page offset stays at chance (roster-mean C3 0.50), and the pairwise test that separates *data artifacts* (S6, which recovers to perfect once the figure is rendered) from genuine blindness leaves G6 on the floor for *every* model. A stress-test addendum identifies the boundary: when the same rigid offset moves content 48 px beyond the page edge, both C0 and C3 detect the defect at 0.958 balanced accuracy (11/12 positives, 12/12 clean controls), although C0 never names G6 while C3 does. Thus the blind spot is genuine over the tested moderate, unclipped range but not magnitude-invariant; overt clipping restores visibility. Declared geometry is still assigned to the linter because it detects both regimes without relying on this visual threshold.

Template snapping can remove injected defects. Enterprise “snap-to-master” templates re-fit element boxes to a grid before rendering, and we find a template silently *absorbs* 45% of injected geometry defects: overlap, alignment, and margin violations become pixel-identical to their clean twins after snapping, while overflow (content-determined) survives, a trap for any dataset that derives labels from the IR but evaluates on the rendered image, quantified and tied to a reward-model failure in Sec. 5.3.

5 From diagnosis to a deterministic symbolic–neural critic

The diagnosis says a single model is the wrong unit. We prescribe a fixed correspondence from diagnosed failure type to inspection method, validate the elicitation recovery across models, and stress the composition with the constructed render class G7. This assignment is deterministic, not learned from class outcomes.

5.1 Elicitation separates format suppression from unresolved capability limits

Four ways to ask. Holding the model and image fixed, we compare four prompting conditions. C0 is one call that scores every defect type against a rubric. C1 asks for a free-form description before classification. C2 uses a synthetic-twin pair in which the clean reference is re-rendered to a grid. C3 issues one yes/no query per defect type and requires localization of the offending element. The primary comparison is C3

³The original S6 render omitted the figure element, so the cross-modal contradiction was literally unseeable (recall 0); on the valid-figure corpus the S6 2-AFC reproduces at 0.75–1.00 across all six VLMs, confirming it is a render artifact, not blindness.

against C0. The model, image, and taxonomy remain fixed; only the whole-rubric versus atomic evidence-bearing format changes. A gain under C3 therefore identifies sensitivity to the elicitation format under this controlled comparison.

C3 recovers G7 within the capable subset. We scope the claim to models that can see G7 at all: a model counts as “capable” only if, under C3, it both detects the overflow (balanced accuracy ≥ 0.70 and precision ≥ 0.70 on paired clean controls) and points at the overflowing element; selected on the same test items, this subset is descriptive rather than confirmatory. This excludes InternVL-8B and Ovis-9B, which stay near chance, a capability boundary rather than a format effect. On the four capable models (three scales, three vendors), C3 recovers what C0 suppresses: balanced accuracy 0.50–0.79 $\rightarrow$ 0.93–1.00, precision 0.88–1.00. The effect appears across models. Pairing each slide across the four models, C3 fixes 233 G7 items that C0 missed against 2 reversals (McNemar $\chi^2_1=227$, $p < 10^{-50}$).⁴ C0’s one-call rubric does not include G7, which is a render class outside the original taxonomy. C3 asks about that class directly. The next controls separate the effect of naming the class from the effect of using an atomic query.

Naming and query format have different effects. Is it naming the class that helps, or changing how we ask? Two controls separate this. C0+ is C0 with G7 added to the rubric’s defect list. It retains the one-call format but gives the model a name for the class. C0_*named* instead asks about G7 alone, on its own yes/no call. Table 2 shows the split. Adding the name to the one-call rubric (C0+) recovers recall (234 defectives newly flagged, $p=7 \times 10^{-71}$), but it reduces specificity: clean slides are over-flagged on 55% of pairs, compared with 2.4% for C3, so balanced accuracy remains lower. Asking about G7 on its own call (C0_*named*) recovers both, and supplying a clean reference slide adds nothing on top. So for G7 the suppressor is the *one-call format*: under a targeted atomic query, the same model on the same image recovers the detection the whole-taxonomy rubric buries. The boundary is G1/S6: naming alone does *not* help. Those classes need the clean reference slide (pairwise choice $\rightarrow$ 0.97–1.00), a *reference-assisted* effect rather than single-slide format suppression. A clean-vs-clean control rules out a forced-pick artifact (capable models answer “tie” on 92–100% of clean pairs).

The recovery persists under compute matching. The remaining alternative is that C3 wins only because it spends more test-time compute than the one-call rubric. We test that directly on three API-served models spanning distinct vendors, holding the image and rendered corpus fixed and matching C3’s budget with repeated C0 draws. The compute-matched control is *self-consistency*: majority vote over $K=10$ independent C0 samples. On G7, C3 still beats that compute-matched control on all three vendors (0.92/0.88/0.83 vs. 0.61/0.59/0.63; $\Delta=+0.31/+0.29/+0.20$), and all three contrasts survive Holm correction within

⁴Per-model $p \leq 1.5 \times 10^{-11}$, all surviving Holm correction over the final paper-wide 85-test family; the two reversals are on Gemma4-31B (C3 recall 88/90). Full table in the supplement.

Table 2: Naming vs. format on G7. Adding the class name to the one-call rubric (C0+) recovers recall but wrecks specificity (55% clean over-flag); asking about G7 on its own call (C0_named, C3) recovers both. C0_named matches C3, so a clean reference adds nothing for G7.

elicitation (G7)	clean over-flag	bal-acc
C0 (whole-tax., unnamed)	2.4%	0.50
C0+ (whole-tax., G7 named)	55%	0.58–0.82
C0_named (atomic, no ref.)	2.4%	0.93–1.00
C3 (atomic, +forced loc.)	2.4%	0.93–1.00

the full 24-test family (reported in the supplement). The budget points the same way: C3 is the cheapest condition in completion tokens (34.7/37.5/40.5 output tok per slide, one call) while the matched self-consistency baseline spends 8–10 calls per slide and still loses. A larger sampling budget therefore does not account for the recovery in this experiment. Changing the whole-taxonomy pointwise format recovers detection without increasing test-time compute. We report the union aggregation of the same draws, the definition-matched one-call control, and the G1 negative control in the supplement.

Elicitation depends on model capability and defect class.

The best elicitation depends jointly on model capability and defect class. No single prompt covers all cases, so inspection must be assigned per diagnosed failure: the C2 synthetic-twin pairwise rescues declared G1 on capable models (snapping absorbs the overflow and the contrast exposes it), while C1 free-form rescues weak models but collapses on strong ones (precision ≈ 0.50). The format effect is not a synthetic artifact: on human-annotated SLIDEAUDIT (no structure) C3 still beats C0 for every class (e.g. brand color, alignment). Absolute geometry on bare real pixels stays modest outside blatant overlap. Real fine geometry needs the linter, hence the structure bare images lack, so on image-only real data the hybrid degrades to VLM-only, a ceiling set by missing structure, not method.

5.2 The minimal hybrid critic covers the most

The assigned methods provide complementary coverage.

On a separate novel-template held-out validation set (150 positives and 150 clean controls per class), the corrected deterministic composition reaches macro balanced accuracy .957 and covers 8/9 classes; the sole miss is S6 (.677 balanced accuracy, .607 precision), whose perfect recall is offset by clean false positives (full table and fidelity audit in the Technical Supplement). On a later disjoint seed-controlled set generated after the method assignment was frozen (30 positives and 30 clean controls per class), the completed frozen image arm scores macro balanced accuracy .826 and named localization .778, with unresolved clean-reference requests counted as misses. These are distinct stages and datasets: the former is held-out validation whose results had already been inspected, while the latter is a separate frozen check rather than a rerun on that corpus. We score each critic by *named attribution*: a detection is correct only if it names

Table 3: Per-class named-attribution balanced accuracy [95% Wilson]; n = paired positives (G1 40, G2 40, G3 70, G5 40, G6 80, G7 40, S1 18, S4 36, S6 12; precision per cell in the Technical Supplement). The minimal linter+VLM-C3 hybrid has the highest mean balanced accuracy. “covered” means bal-acc > 0.75 and Wilson lower bound > 0.65 ; under that rule, linter+VLM-C3 and the prescribed composition each clear 6/9 classes. The prescribed column reports the corrected S1 method; the frozen text-only S1 baseline (0.25, precision 0.09) is reported in the prose below. Small S1/S6 cells are method-selection diagnostics, not confirmatory (those claims rest on the elicitation decomposition and 2-AFC).

Class	linter	VLM-C0	VLM-C3	linter+C3	prescribed
G1 overflow	1.00 [.91,1]	0.50 [.46,.54]	0.50 [.46,.54]	1.00 [.91,1]	1.00 [.91,1]
G2 overlap	0.80 [.68,.87]	0.71 [.60,.79]	0.86 [.74,.92]	0.80 [.68,.87]	0.80 [.68,.87]
G3 alignment	0.93 [.85,.96]	0.62 [.55,.68]	0.71 [.63,.77]	0.93 [.85,.96]	0.93 [.85,.96]
G5 color	1.00 [.91,1]	0.50 [.46,.54]	0.68 [.57,.75]	1.00 [.91,1]	1.00 [.91,1]
G6 margin	0.75 [.67,.80]	0.51 [.40,.61]	0.51 [.47,.55]	0.75 [.67,.80]	0.75 [.67,.80]
G7 render-overflow	0.00 [0,.54]	0.50 [.46,.54]	1.00 [.91,1]	1.00 [.91,1]	1.00 [.91,1]
S1 title/body	0.50 [.41,.59]	0.94 [.75,.98]	0.83 [.63,.92]	0.83 [.63,.92]	0.94 [.75,.98]
S4 density	0.50 [.45,.55]	0.51 [.44,.59]	0.93 [.81,.97]	0.93 [.81,.97]	0.72 [.60,.80]
S6 image/text	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]
mean / covered	0.66 / 4/9	0.59 / 1/9	0.72 / 3/9	0.86 / 6/9	0.85 / 6/9

the right defect type on the defective slide and stays silent on the matched clean slide. We call a class “covered” when this balanced accuracy exceeds 0.75 (with the 95% Wilson lower bound above 0.65, so the cell is not just lucky). Coverage tracks the diagnosis: declared geometry is covered by the linter, the format-suppressed classes by a C3-elicited VLM. The mean balanced accuracy summarizes this complementary coverage. The minimal hybrid of a linter and one C3-elicited VLM reaches 0.86, compared with 0.59 for the conventional pointwise VLM-C0 and 0.66 for the linter alone (Table 3). Under the same strict covered criterion, the minimal hybrid clears 6 of the 9 image-level classes; G6 at 0.75 [.67,.80] misses by the paper’s own “exceeds” rule, and S1 under linter+C3 misses because its lower bound stays below 0.65. The nine classes are the image-level ones; G4/S3 are linter-owned and S2/S5 are deck-scope (Sec. 6). Note that this assignment needs no per-class tuning: the rule follows from the diagnosis, not from optimizing over class outcomes, so the coverage table is what the diagnosis predicts rather than a tuned architecture.

The S1 diagnosis changes its assignment. We initially assigned S1 title/body mismatch to the text LLM, but the text-only probe over-flags (0.25, precision 0.09) while the image-bearing VLM names it reliably (0.94, precision 0.90): title/body mismatch needs the *rendered layout*, not just the text. We therefore report both numbers: the frozen text-only baseline and the corrected VLM method. Under the corrected assignment, S1 uses the VLM under its open pointwise prompt (C0, 0.94) rather than C3 (0.83), since recall is already saturated and C3 only adds false positives. This case shows how the diagnosis changes method selection. The one image-level class the hybrid does not cover is S6 im-

age/text contradiction, at 0.50 single-image for every engine; the pairwise test shows the model *does* perceive it (recall 1.0, specificity ~ 0 , over-asserting rather than missing), so this is a reference-assisted effect, not a blind spot, in contrast to the moderate, unclipped G6 page-offset regime.

5.3 G7: the linter-blind class that specialised rewards also miss

Construction and scorer response. G7 makes the linter’s blind spot falsifiable: a declared box that is legal (in-margin, non-overlapping) but whose rendered content overflows its container. The IR carries only legal short text; the overflow lives in the renderer, so the linter and any structure-only model are blind by construction. A self-check confirms zero findings on 90/90 G7 defectives. We then audit five published scorers plus one frontier VLM judge on the same paired slides. The same-backbone comparison provides the most direct separation: CLIP-IQA detects G7, whereas the LAION aesthetic head on the same CLIP-L/14 backbone is at chance. Skywork, PickScore, and Qwen3.7-Max also detect the class, whereas DocReward does not (full per-scorer table in the Technical Supplement). So on the scorers tested, G7 tracks a rendered-quality read-out rather than the training-domain label.

Template snapping removes 45% of injected geometry. The reward audit ties to the template result of Sec. 4. Rendering every injected defective both faithfully and snapped-to-master, 45% of injected geometry defects (100% of overlap, alignment, margin; 20% of overflow) become pixel-identical to their clean twin after snapping. Feeding the *snapped* pairs to all five scorers gives preference accuracy 0.00 at a reward gap of exactly 0.000: identical inputs force identical scores. This is a dataset-integrity hazard, not a reward-model deficiency, and it is model-agnostic by construction: any reward model trained or evaluated on snap-rendered images with IR-derived labels inherits these zero-signal pairs. Perturbation fidelity must be checked at the pixel level, not assumed from the IR.

6 The semantic engine: a specialist examiner

The hybrid’s semantic engine can be a zero-shot model, but a small specialist does better on its own ground. We finetune Qwen3-VL-8B with QLoRA on 2,142 injected supervision records; the training mix is summarized in the Technical Supplement.

The 8B examiner exceeds a zero-shot 30B on in-distribution semantics. On in-domain held-out page-semantic inspection, the finetuned 8B reaches balanced accuracy 1.000 on both S1 title/body and S4 density, beating both zero-shot baselines including the 30B. On these in-distribution semantic tasks, the specialist 8B exceeds the larger zero-shot model. The largest gap is S4 density (1.00 vs. 0.77 for the 30B), and both in-distribution contrasts survive Holm correction over the final 85-test family. Full format-control and severity breakdowns are in the Technical Supplement.

The examiner abstains on geometry. Under image-only inspection, the finetuned model stays at 0.50 on G2–G6 with *zero* false positives: trained with image-only abstention targets, it declines to hallucinate geometry from pixels and leaves those classes to the linter, the designed behavior, and the reason the hybrid’s geometry column comes from the linter rather than the VLM.

The examiner does not transfer to bare real pixels. The examiner does not transfer as a bare-pixel real-world inspector. On image-only SLIDEAUDIT, its synthetic S4 strength collapses to chance (0.50), whereas only the zero-shot 30B shows real density signal ($p=1.1 \times 10^{-5}$). Scale, not fine-tuning, carries bare-pixel transfer: the examiner learned a distribution, not universal slide quality.

7 External validity

A real pre-sales agent deck exposes distribution shift. On a real 16-slide proposal from a deployed PowerPoint agent, the verifiable defect is 20 unfilled template placeholders (“add a title here”) across 7 slides. This semantic-completeness defect is outside the linter’s rules. The zero-shot 30B is the best detector (recall 1.0, names the placeholder on 4/7 slides, scores them 0.03 vs. 0.64 for clean), while the *finetuned* 8B, top of the intrinsic ranking, is among the *worst* here (0/7), and the linter’s apparent recall is spurious geometry noise: on a novel defect the ranking is by general-VLM ability rather than specialist in-distribution quality. The examiner learned a distribution, not universal quality. Feeding the flagged placeholders back to the generator for one revision drives the defect from 20 to 0.

Real layouts reproduce the attribution outcomes. The preceding A/B/C attribution uses synthetic slides. We repeat it on *real* geometry with a declared structural oracle and obtain the oracle without human or model annotation: ZENODO10K (Zheng et al. 2025) ships real CC-licensed .pptx with intact element XML, so `python-pptx` extracts a geometry/text/style oracle *lossless with respect to the declared IR* (the IR $\rightarrow$ render gap that G7 exploits is exactly what this oracle does *not* capture). From 26 real decks we inject one single-shape defect per slide in *PPTX XML space* (overflow, overlap, alignment, font, margin) and render both the clean and defective one-slide deck with the same renderer (LibreOffice), so the paired pixel diff isolates the injected defect on real pixels (209 paired slides, 5 classes, 3 VLMs / 3 families). Two results follow. First, the §4 “snap absorbs 45%” hazard is *template-specific*: real free-form decks absorb only 7% (render-fidelity 0.93), a positive control. Second, *all three* attribution outcomes appear on real geometry: coarse defects (overflow, overlap) are image-sufficient (A 0.70–0.74, structure adds nothing); a margin bleed is a perception bottleneck the oracle rescues for weaker VLMs (A 0.59 $\rightarrow$ B 0.70); and fine alignment stays near chance in *every* channel for *every* model (A=B=C $\approx$ 0.50; a supra-threshold 44 px internal-contrast re-run reproduces this, A/B/C 0.51–0.54, $n=41$). This is *not* a hard capability floor (on clean slides the same defect recovers under targeted elicitation once it clears acuity), but on real cluttered layouts the model does not isolate

the misaligned element reliably among genuine layout variation, and specificity collapses, so for declared geometry the symbolic linter remains the reliable detector regardless.

Pixel-recovered structure does not restore the linter. A third-party PDF/PNG deck ships no IR, so we ask whether *recovering* element structure from pixels re-arms the linter. Following DocLayNet/PubLayNet (Pfitzmann et al. 2022; Zhong, Tang, and Yepes 2019), PP-DocLayoutV2 returns region boxes per render (class-agnostic non-maximum suppression removes cross-class duplicates) that feed the *same* linter. The answer is a pre-registered negative: recovered structure leaves the linter at chance on G1/G3/G4 and rescues only overlap partially (G2 0.54 vs. native-IR 0.67 vs. VLM-only 0.87); the detector *over-segments* ($\approx 2\times$ more boxes, recovery recall at intersection-over-union (IoU) ≥ 0.5 only 0.30–0.39), so collisions split and overlap recall falls $0.78 \rightarrow 0.29$. On image-only SLIDEAUDIT the recovered linter clears chance only on G2 (0.55) while the C3-elicited VLM reaches 0.94. On real third-party decks even *native* IR is weakened (no position/font bookkeeping, so G3/G4 rules are at chance and margin/overlap rules *over-fire*, specificity 0.11/0.56), so the VLM-only engine has higher balanced accuracy on every coarse class in this setting. The linter’s advantage is limited to in-distribution, clean native IR; with open-world inputs, the deployable critic remains bounded by the VLM.

8 Limitations

We now report a human spot-check on the rebuilt corpus, but not an inter-annotator or deployment-time human baseline. On the rebuilt v2 sample, every sampled defect is judged visible and every clean twin clean (73/73; 55/55 source-faithful on the auditable structure-carrying subset), so the paper’s use of G3/G5 rests on corrected human-verified samples rather than the stale legacy framing. Three scope bounds remain. First, the real-layout attribution (§7) uses *injected* defects, so image-only SLIDEAUDIT is the complementary real probe. Second, the reward-audit document-reward miss is instance-level, so the categorical read rests on the two general-multimodal backbones and the same-backbone CLIP-IQA/LAION dissociation. Third, the examiner $\rightarrow$ generation downstream effect is near-null: quality stays at $\Delta q = +0.00$ where a coverage critique moves it by $+0.32$. The contribution is the *diagnosis* and deterministic method prescription, not a claim to match human inspectors.

9 Conclusion

A slide-defect critic should not be one model. Failure attribution prescribes deterministic composition: declared geometry goes to a symbolic linter, format suppression to targeted single-slide elicitation, comparison-resolvable defects to paired references, and page semantics to a semantic text/examiner module. Atomic recovery beats compute-matched controls without extra inference. The composition reaches .957 on held-out validation; a later frozen image-arm check scores .826 with unresolved reference requests counted as misses. S6 clean false positives, failed pixel-recovered structure, and the SLIDEAUDIT transfer gap delimit the result.

Reproducibility: code, linters, harness, and reward adapters are in the Code and Data Supplement; family-size updates are reported there.

References

- Fu, S.; Bonnen, T.; Guillory, D.; and Darrell, T. 2025. Hidden in Plain Sight: VLMs Overlook Their Visual Representations. arXiv:2506.08008.
- Ge, J.; Wang, Z. Z.; Zhou, X.; Peng, Y.-H.; Subramanian, S.; Tan, Q.; Sap, M.; Suhr, A.; Fried, D.; Neubig, G.; and Darrell, T. 2025. AutoPresent: Designing Structured Visuals from Scratch. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Heo, I.; Hwang, T.; Jung, J.; and Jung, S. 2025. LED Benchmark: Diagnosing Structural Layout Errors for Document Layout Analysis. arXiv:2507.23295.
- Kang, H.; Bao, E.; and Goswami, A. 2025. VLM-SlideEval: Evaluating VLMs on Structured Comprehension and Perturbation Sensitivity in PPT. arXiv:2510.22045.
- Kumar, D.; Tayebati, S.; Naik, D.; Krishnan, R.; and Trivedi, A. R. 2026. VLM Judges Can Rank but Cannot Score: Task-Dependent Uncertainty in Multimodal Evaluation. arXiv:2604.25235.
- Liu, C.; Yang, Y.; Zhou, K.; Zhang, Z.; Fan, Y.; Xie, Y.; Qi, P.; and Wang, X. E. 2025a. Presenting a Paper is an Art: Self-Improvement Aesthetic Agents for Academic Presentations (EvoPresent/PresAesth). arXiv:2510.05571.
- Liu, J.; Zeng, W.; Zhang, X.; Wang, Y.; Shan, Z.; and He, J. 2025b. On the Perception Bottleneck of VLMs for Chart Understanding. arXiv:2503.18435.
- Liusie, A.; Manakul, P.; and Gales, M. 2024. LLM Comparative Assessment: Zero-shot NLG Evaluation through Pairwise Comparisons using Large Language Models. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*.
- Pan, Y.; Hu, C.; Huang, X.; Huang, C.; Zhao, M.; Bi, Y.; Zhang, X.; Zeng, A.; and Hu, L. 2026. AeSlides: Incentivizing Aesthetic Layout in LLM-Based Slide Generation via Verifiable Rewards. arXiv:2604.22840.
- Pfitzmann, B.; Auer, C.; Dolfi, M.; Nassar, A. S.; and Staar, P. W. J. 2022. DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.
- Tong, S.; Liu, Z.; Zhai, Y.; Ma, Y.; LeCun, Y.; and Xie, S. 2024. Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Tripathi, T.; Wadhwa, M.; Durrett, G.; and Niekum, S. 2025. Pairwise or Pointwise? Evaluating Feedback Protocols for Bias in LLM-Based Evaluation. arXiv:2504.14716.
- Wang, X.; Huang, J.; Zhang, X.; Wang, T.; and Ma, J. W. 2025. Your Reasoning Benchmark May Not Test Reasoning: Revealing Perception Bottleneck in Abstract Reasoning Benchmarks. arXiv:2512.21329.

Zhang, Z. J.; Chen, R.; Zhong, M.; and Wobbrock, J. O. 2025. SlideAudit: A Dataset and Taxonomy for Automated Evaluation of Presentation Slides. arXiv:2508.03630.

Zheng, H.; Guan, X.; Kong, H.; Zheng, J.; Zhou, W.; Lin, H.; Lu, Y.; He, B.; Han, X.; and Sun, L. 2025. PPTAgent: Generating and Evaluating Presentations Beyond Text-to-Slides. Introduces the Zenodo10K corpus of real .pptx; dataset Forceless/Zenodo10K, arXiv:2501.03936.

Zhong, X.; Tang, J.; and Yepes, A. J. 2019. PubLayNet: Largest Dataset Ever for Document Layout Analysis. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR)*.